



North Sydney Girls High School

2026

HSC TRIAL EXAMINATION

# Mathematics Extension 1

## General Instructions

- Reading Time – 10 minutes
- Working Time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations

**Total marks:**  
**70**

### Section I – 10 marks (pages 2 – 7)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

### Section II – 60 marks (pages 8 – 15)

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

NAME: \_\_\_\_\_

TEACHER: \_\_\_\_\_

STUDENT NUMBER:

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| Question | 1-10 | 11  | 12  | 13  | 14  | Total |
|----------|------|-----|-----|-----|-----|-------|
| Mark     | /10  | /15 | /15 | /15 | /15 | /70   |

## Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for Questions 1-10.

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1 Given that  $\overrightarrow{OA} = \begin{pmatrix} 2 \\ -7 \end{pmatrix}$  and  $\overrightarrow{OB} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$ , what is  $\overrightarrow{AB}$ ?

A.  $\begin{pmatrix} 5 \\ 11 \end{pmatrix}$

B.  $\begin{pmatrix} -5 \\ 11 \end{pmatrix}$

C.  $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$

D.  $\begin{pmatrix} -1 \\ -3 \end{pmatrix}$

2 The polynomial  $p(x) = x^3 - 6x^2 + 32$  has one repeated zero of multiplicity 2 at  $x = \alpha$ .  
What is the value of  $\alpha$ ?

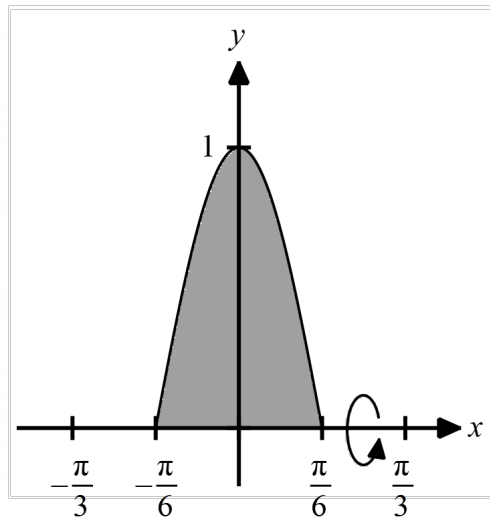
A. -2

B. 0

C. 2

D. 4

- 3 A solid of revolution is to be formed by rotating the region bounded by the  $x$ -axis and the curve  $y = \cos 3x$  between  $x = -\frac{\pi}{6}$  and  $x = \frac{\pi}{6}$  about the  $x$ -axis, as shown in the diagram.

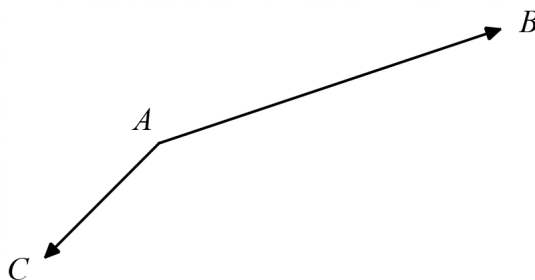


Which of the following gives the volume of solid of revolution formed?

- A.  $\pi \int_0^{\frac{\pi}{6}} 1 + \cos 6x \, dx$
- B.  $\pi \int_0^{\frac{\pi}{6}} \cos^2 3x \, dx$
- C.  $\frac{\pi}{2} \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \cos^2 3x \, dx$
- D.  $\frac{\pi}{2} \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} 1 - \cos 6x \, dx$

- 4 The projection of  $\underline{u}$  onto  $\underline{v}$  is given by  $\left( \frac{\underline{u} \cdot \underline{v}}{|\underline{v}|^2} \right) \underline{v}$ .

The diagram below shows points  $A$ ,  $B$  and  $C$ , where  $\overrightarrow{AB} = 3\hat{i} + \hat{j}$ ,  $\overrightarrow{AC} = -\hat{i} - \hat{j}$ .



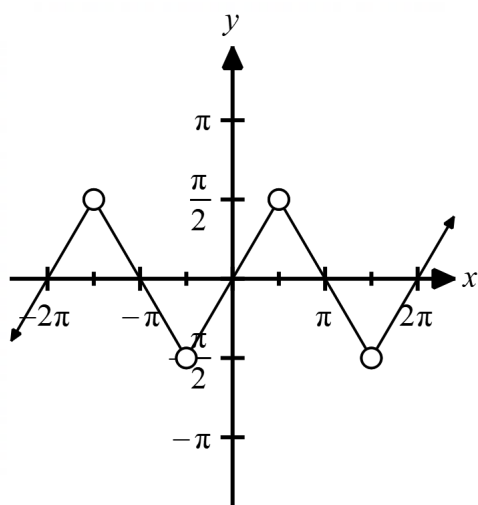
Which of the following is the projection of  $\overrightarrow{AB}$  onto  $\overrightarrow{AC}$ ?

- A.  $\frac{2}{5}(3\hat{i} + \hat{j})$
- B.  $\frac{-2}{5}(3\hat{i} + \hat{j})$
- C.  $2(\hat{i} + \hat{j})$
- D.  $2(-\hat{i} - \hat{j})$

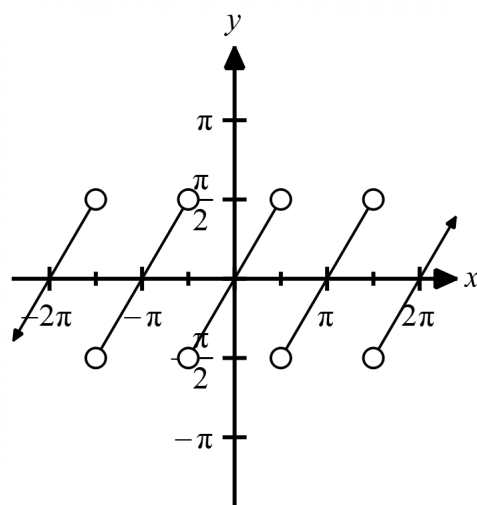
- 5 Five different schools send two prefects each to attend a conference. The group of 10 is to be seated at a circular table. In how many ways can the prefects be seated if prefects from the same school are to be seated together?
- A.  $4!$
- B.  $2 \times 4!$
- C.  $2^4 \times 4!$
- D.  $2^5 \times 4!$
- 6 You are given that  $f(x)$  is a one-to-one function defined for  $x \in (-\infty, \infty)$ . Which of the following functions cannot have an inverse?
- A.  $f(|x|)$
- B.  $|f(x)|$
- C.  $\frac{1}{f(x)}$
- D.  $\sqrt{f(x)}$
- 7 How many distinct solutions are there to the equation  $\sin x + \sin 3x = \sin 2x$  for  $0 \leq x \leq 2\pi$ ?
- A. 5
- B. 6
- C. 7
- D. 8

8 Which graph represents the function  $y = \tan^{-1}(\tan x)$ ?

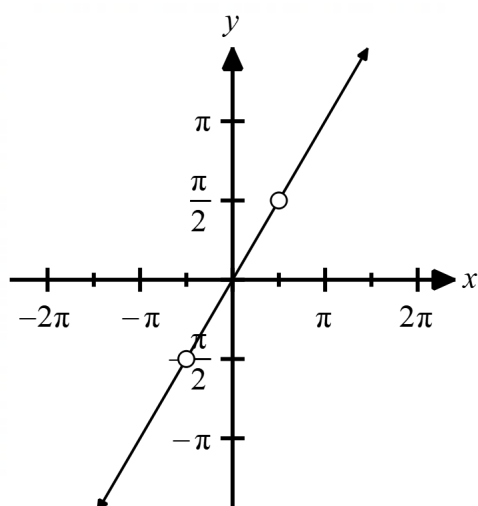
A.



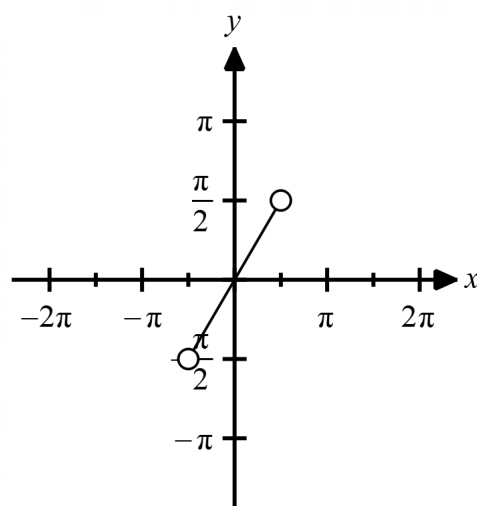
B.



C.



D.



- 9 It is known that the function  $f(x)$  is one-to-one and continuous for all real  $x$ .

Let  $g(x) = f^{-1}(x)$ .

Which of the following statements MUST be true for any value of  $x$  where  $f'(x)$  and  $g'(x)$  are both defined?

A.  $f'(x) = \frac{-1}{g'(x)}$

B.  $f'(x) = \frac{1}{g'(x)}$

C.  $f'(x)g'(x) < 0$

D.  $f'(x)g'(x) > 0$

- 10 Which statement is always true for real numbers  $a$  and  $b$  where  $0 \leq a < b \leq 1$ ?

A.  $\cos^{-1} a < \cos^{-1} b$

B.  $\cos^{-1} a - \sin^{-1} a > \cos^{-1} b - \sin^{-1} b$

C.  $\cos^2 a < \cos^2 b$

D.  $\sin a \cos a > \sin b \cos b$

**End of Section I**

## Section II

**60 marks**

**Attempt Questions 11-14**

**Allow about 1 hour and 45 minutes for this section**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

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**Question 11** (15 marks) Use a SEPARATE writing booklet

- (a) The parametric equations of a line are given below. **2**

$$x = 2 - 5t$$

$$y = 3t + 1$$

Find the Cartesian equation of this line in the form  $y = mx + c$ .

- (b) Find the inverse function,  $f^{-1}(x)$ , of the function  $f(x) = e^{2x} - 1$ . **2**

- (c) The graph of  $y = f(x)$  is given on the separate Response Sheet for Question 11 (c). **2**

On the graph provided in the response sheet, sketch the graph of  $y = \sqrt{f(x)}$ , clearly showing all important features.

- (d) Consider the polynomial **3**

$$P(x) = x^3 - 2x^2 - 7x - 4.$$

Show that  $x + 1$  is a factor of  $P(x)$  and hence find  $P(x)$  in fully factorised form.

**Question 11 continues on page 9**

Question 11 (continued)

(e) Solve  $\frac{2x}{3-x} \leq 4$ . **3**

(f) Find the coefficient of  $x^{10}$  in the expansion of  $\left(\frac{2}{x} - x^3\right)^6$ . **3**

**End of Question 11**

**Question 12** (15 marks) Use a SEPARATE writing booklet

(a) Use Mathematical induction to prove that  $4n^3 - n$  is divisible by 3 for all integers  $n \geq 1$ . **3**

(b) (i) Express  $3 \sin x - \sqrt{3} \cos x$  in the form  $2\sqrt{3} \sin(x + \alpha)$ , where  $0 < \alpha < 2\pi$ . **2**

(ii) Hence, or otherwise, solve  $3 \sin x = \sqrt{3} \cos x - \sqrt{3}$ , where  $0 \leq x < 2\pi$ . **2**

(c) Evaluate  $\int_0^{\frac{\sqrt{3}}{2}} \frac{8x^2}{\sqrt{1-x^2}} dx$  using the substitution  $x = \sin \theta$ . **4**

(d) Charles is a member of an elite racing group of 11 people. A team of 5 people is to be formed from the group. Use a combinatorial argument in this context to prove that **2**

$$\binom{11}{5} = \binom{10}{4} + \binom{10}{5}.$$

(e) By sketching a graph, or otherwise, solve  $(|x| - 1)(|x| - 2) = 6$ . **2**

**End of Question 12**

**Question 13** (15 marks) Use a SEPARATE writing booklet

(a) Let  $f(x) = \tan^{-1} x - \tan^{-1} \left( \frac{x-1}{x+1} \right)$ , for  $x > -1$ .

(i) Show that  $f(x) = k$  where  $k$  is a real number. 2

(ii) Hence, find the value of  $k$ . 1

(b) A particle is moving on the curve with equation 4

$$y = 2 \arcsin 3x, \quad -\frac{1}{3} \leq x \leq \frac{1}{3}.$$

The particle has coordinates  $(x, y)$  at time  $t$ , where distances are measured in metres and time in seconds.

At  $t = T$ , the  $y$  coordinate of the particle is  $\frac{1}{3}\pi$  m and the  $y$  coordinate is increasing at a rate of 2 m/s.

Find the exact rate at which the  $x$  coordinate of the particle is changing at  $t = T$ .

(c) At a particular high school, 17 students, including Xavi and Yuri, are enrolled to study Mathematics Extension 1. There are three Mathematics Extension 1 classes with a maximum class size of 6. Each class is taught by a different teacher.

(i) Will there always be a class that is full? Justify your answer. 1

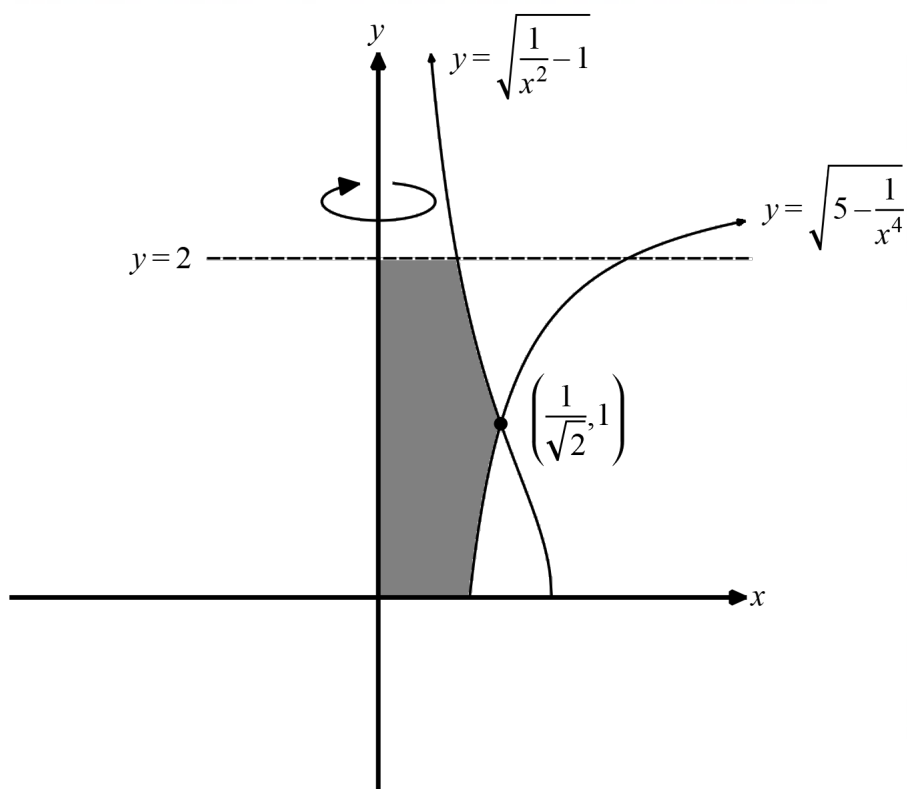
(ii) Students are randomly assigned into classes. Find the number of ways the classes can be assigned to teachers such that Xavi and Yuri are in the same class. 2

**Question 13 continues on page 12**

Question 13 (continued)

- (d) A solid trophy is to be made by rotating the shaded region below around the  $y$ -axis.

5



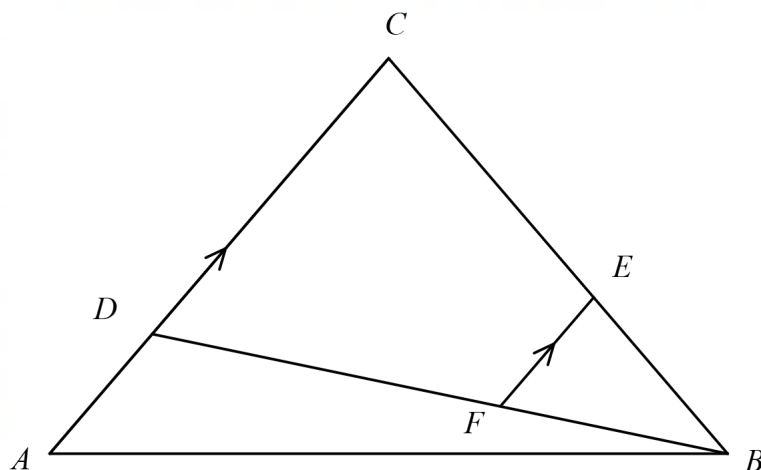
Show that the volume of the trophy is  $\frac{\pi^2}{4} u^3$ .

**End of Question 13**

**Question 14** (15 marks) Use a SEPARATE writing booklet

- (a) In triangle  $ABC$ ,  $D$  lies on  $AC$  such that  $\overrightarrow{AD} = \frac{1}{4}\overrightarrow{AC}$ ,  $E$  lies on  $BC$  such that  $\overrightarrow{BE} = \frac{1}{3}\overrightarrow{BC}$ .

$F$  lies on  $BD$  such that  $FE \parallel AC$ .



- (i) Show that  $\overrightarrow{FE} = \overrightarrow{FB} + \frac{1}{3}\overrightarrow{BC}$ . **1**
- (ii) Let  $\overrightarrow{FE} = k\overrightarrow{DC}$ . Use vector methods to show that  $\overrightarrow{FE} = \overrightarrow{AD}$ . **3**

**Question 14 continues on page 14**

Question 14 (continued)

- (b) The polynomial  $p(x) = x^3 + bx^2 + cx + d$  has stationary points at  $x = \alpha$  and  $x = \beta$ , where  $\alpha \neq \beta$ .

(i) Show that  $b^2 > 3c$ . **2**

(ii) You are given that  $m^3 - n^3 = (m - n)(m^2 + mn + n^2)$ . (Do NOT prove this) **3**

Show that  $\frac{p(\alpha) - p(\beta)}{\alpha - \beta} = \frac{2}{9}(3c - b^2)$ .

(iii) Hence, or otherwise, find the relationship between  $\alpha$  and  $\beta$  such that  $p(\alpha) > p(\beta)$ . **1**

**Question 14 continues on page 15**

Question 14 (continued)

- (c) Particle  $A$  is projected from the origin with initial speed  $V \text{ m s}^{-1}$  at an angle  $\theta$  with the horizontal plane. The position vector of particle  $A$ ,  $t$  seconds after it is projected, is given by

$$\vec{r}_A(t) = \begin{pmatrix} Vt \cos \theta \\ Vt \sin \theta - \frac{1}{2}gt^2 \end{pmatrix}. \quad (\text{Do NOT prove this})$$

- (i) Show that the velocity vector of particle  $A$  is given by  $\vec{v}_A(t) = \begin{pmatrix} V \cos \theta \\ V \sin \theta - gt \end{pmatrix}$ . 1

After  $T$  seconds, particle  $B$  is projected from the origin with the same initial speed and angle as particle  $A$ . The position vector of particle  $B$  is given by

$$\vec{r}_B(t) = \begin{pmatrix} V(t-T) \cos \theta \\ V(t-T) \sin \theta - \frac{1}{2}g(t-T)^2 \end{pmatrix}, \quad t \geq T. \quad (\text{Do NOT prove this})$$

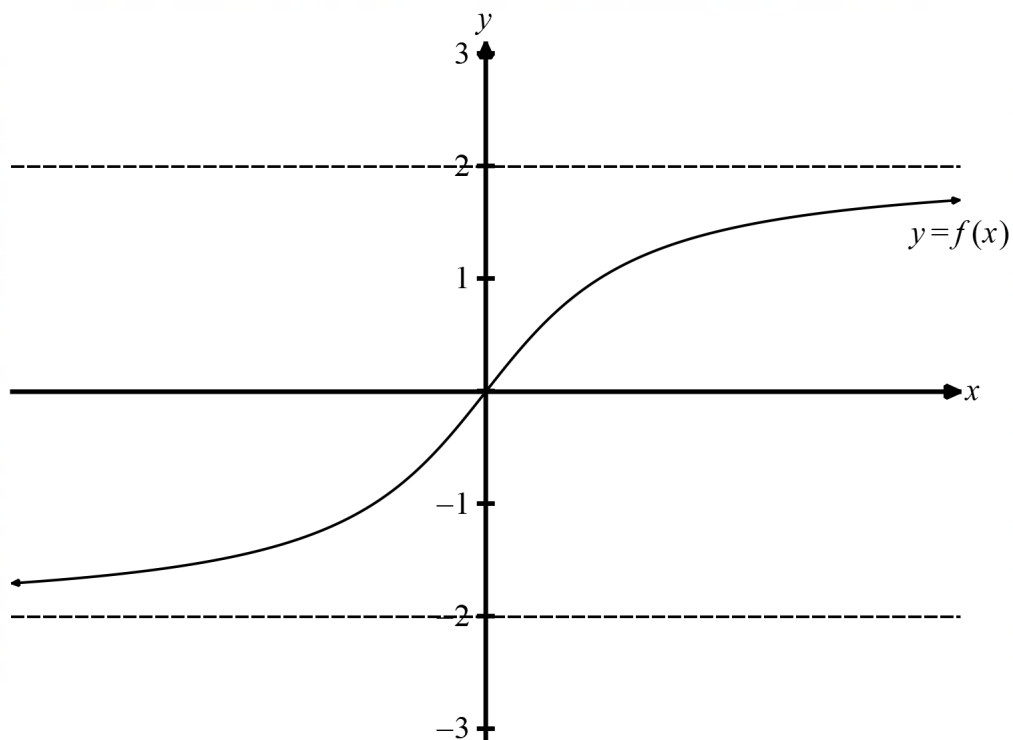
- (ii) Show that the particles will never travel in perpendicular directions at any time  $t$  if  $T < \frac{2V \cos \theta}{g}$ . 4

**End of paper**

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**Question 11 (c) – Response Sheet**

11 (c) The graph of  $y = f(x)$  is shown below.



On the same grid above, sketch the graph of  $y = \sqrt{f(x)}$ , clearly showing all important features.

2

**Place this sheet inside your Question 11 answer booklet.**



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2026

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# Mathematics Extension 1

## General Instructions

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**Total marks:**  
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- Attempt Questions 1 – 10
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### Section II – 60 marks (pages 8 – 15)

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- Allow about 1 hour and 45 minutes for this section

NAME: \_\_\_\_\_

TEACHER: \_\_\_\_\_

STUDENT NUMBER:

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| Question | 1-10 | 11  | 12  | 13  | 14  | Total |
|----------|------|-----|-----|-----|-----|-------|
| Mark     | /10  | /15 | /15 | /15 | /15 | /70   |

## Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for Questions 1-10.

---

1 Given that  $\overrightarrow{OA} = \begin{pmatrix} 2 \\ -7 \end{pmatrix}$  and  $\overrightarrow{OB} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$ , what is  $\overrightarrow{AB}$ ?

A.  $\begin{pmatrix} 5 \\ 11 \end{pmatrix}$

B.  $\begin{pmatrix} -5 \\ 11 \end{pmatrix}$

C.  $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$

D.  $\begin{pmatrix} -1 \\ -3 \end{pmatrix}$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} -3 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ -7 \end{pmatrix} = \begin{pmatrix} -5 \\ 11 \end{pmatrix}$$

2 The polynomial  $p(x) = x^3 - 6x^2 + 32$  has one repeated zero of multiplicity 2 at  $x = \alpha$ . What is the value of  $\alpha$ ?

A. -2

B. 0

C. 2

D. 4

$$p(\alpha) = 0 \text{ and } p'(\alpha) = 0$$

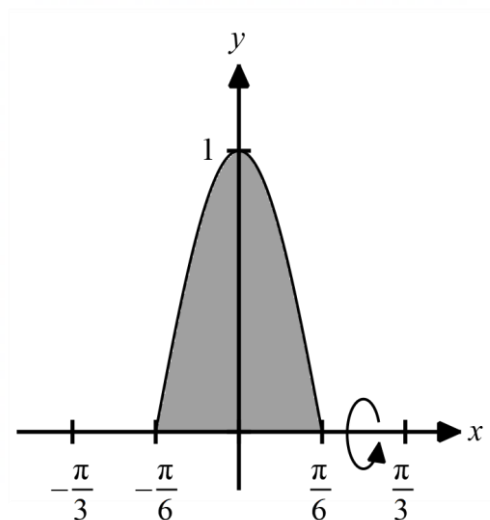
$$p'(x) = 3x^2 - 12x = 3x(x - 4)$$

$$\therefore p'(4) = 0 \text{ and } p'(0) = 0$$

$$p(4) = 4^3 - 6(4)^2 + 32 = 64 - 96 + 32 = 0$$

$$p(0) = 32 \neq 0$$

- 3 A solid of revolution is to be formed by rotating the region bounded by the  $x$ -axis and the curve  $y = \cos 3x$  between  $x = -\frac{\pi}{6}$  and  $x = \frac{\pi}{6}$  about the  $x$ -axis, as shown in the diagram.



Which of the following gives the volume of solid of revolution formed?

A.  $\pi \int_0^{\frac{\pi}{6}} 1 + \cos 6x \, dx$

B.  $\pi \int_0^{\frac{\pi}{6}} \cos^2 3x \, dx$

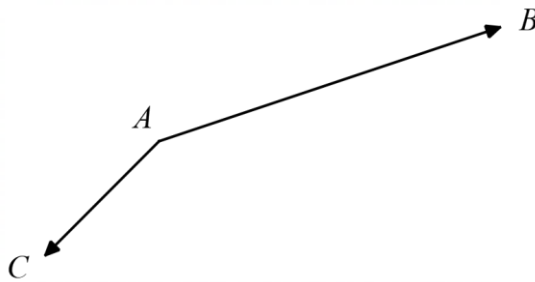
C.  $\frac{\pi}{2} \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \cos^2 3x \, dx$

D.  $\frac{\pi}{2} \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} 1 - \cos 6x \, dx$

$$\begin{aligned}
 V &= \pi \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \cos^2 3x \, dx = 2\pi \int_0^{\frac{\pi}{6}} \cos^2 3x \, dx \quad (\because y = \cos^2 3x \text{ is even}) \\
 &= 2\pi \int_0^{\frac{\pi}{6}} \frac{1}{2} (1 + \cos 6x) \, dx \\
 &= \pi \int_0^{\frac{\pi}{6}} 1 + \cos 6x \, dx
 \end{aligned}$$

- 4 The projection of  $\underline{u}$  onto  $\underline{v}$  is given by  $\left( \frac{\underline{u} \cdot \underline{v}}{|\underline{v}|^2} \right) \underline{v}$ .

The diagram below shows points  $A$ ,  $B$  and  $C$ , where  $\overrightarrow{AB} = 3\hat{i} + \hat{j}$ ,  $\overrightarrow{AC} = -\hat{i} - \hat{j}$ .



Which of the following is the projection of  $\overrightarrow{AB}$  onto  $\overrightarrow{AC}$ ?

- A.  $\frac{2}{5}(3\hat{i} + \hat{j})$
- B.  $\frac{-2}{5}(3\hat{i} + \hat{j})$
- C.  $2(\hat{i} + \hat{j})$
- D.  $2(-\hat{i} - \hat{j})$

Quick check: projection vector is parallel to  $\overrightarrow{AC}$  but in opposite direction to  $\overrightarrow{AC}$ . Only C can be correct.

$$\left( \frac{\underline{u} \cdot \underline{v}}{|\underline{v}|^2} \right) \underline{v} = \left( \frac{3 \times (-1) + 1 \times (-1)}{\left( \sqrt{(-1)^2 + (-1)^2} \right)^2} \right) \overrightarrow{AC} = \frac{-4}{2}(-\hat{i} - \hat{j}) = 2(\hat{i} + \hat{j})$$

- 5 Five different schools send two prefects each to attend a conference. The group of 10 is to be seated at a circular table. In how many ways can the prefects be seated if prefects from the same school are to be seated together?

- A.  $4!$
- B.  $2 \times 4!$
- C.  $2^4 \times 4!$
- D.  $2^5 \times 4!$

Group the pairs of prefects from the same school. There are five groups to arrange in a circular table in  $4!$  ways. Each pair within the group can be arranged in  $2! = 2$  ways for a total of  $2^5$  ways. Therefore, total number of arrangements is  $2^5 \times 4!$ .

- 6 You are given that  $f(x)$  is a one-to-one function defined for  $x \in (-\infty, \infty)$ . Which of the following functions cannot have an inverse?

- A.  $f(|x|)$
- B.  $|f(x)|$
- C.  $\frac{1}{f(x)}$
- D.  $\sqrt{f(x)}$

$f(|x|)$  is always even. Therefore, it cannot have an inverse (its inverse will not be a function)

7 How many distinct solutions are there to the equation  $\sin x + \sin 3x = \sin 2x$  for  $0 \leq x \leq 2\pi$ ?

A. 5

B. 6

C. 7

D. 8

Using sum-to-product result

$$\frac{1}{2}(\sin x + \sin 3x) = \frac{1}{2}(\sin(2x + x) + \sin(2x - x)) = \sin 2x \cos x$$

$$\sin x + \sin 3x = \sin 2x$$

$$2 \sin 2x \cos x = \sin 2x$$

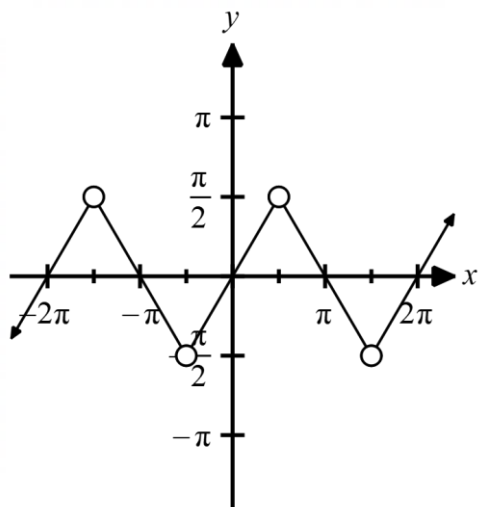
$$\sin 2x(2 \cos x - 1) = 0$$

$$\sin 2x = 0 \quad \cos x = \frac{1}{2}$$

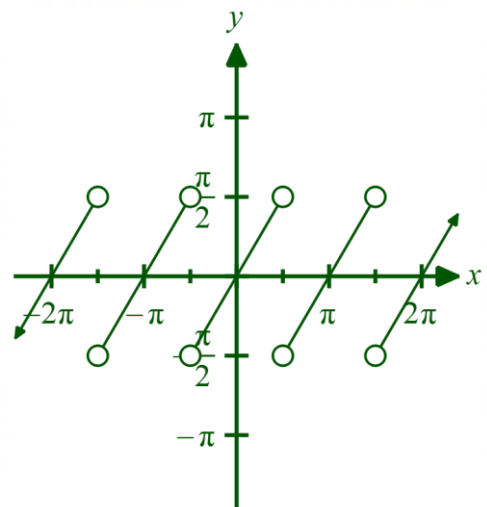
$\sin 2x = 0$  has 5 solutions for  $0 \leq x \leq 2\pi$  and  $\cos x = \frac{1}{2}$  has 2 solutions for  $0 \leq x \leq 2\pi$ , and the solutions are unique. Therefore, there are 7 solutions in total.

8 Which graph represents the function  $y = \tan^{-1}(\tan x)$ ?

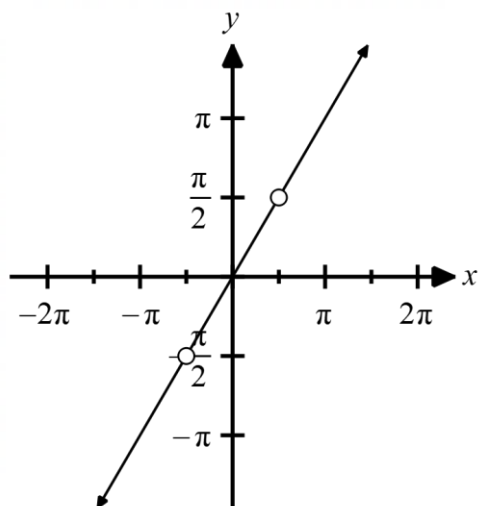
A.



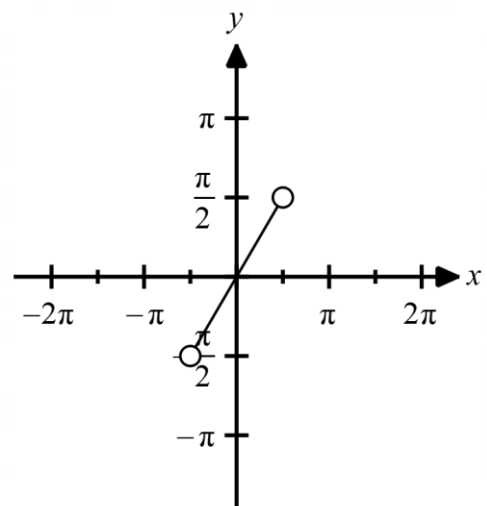
B.



C.



D.



- 9 It is known that the function  $f(x)$  is one-to-one and continuous for all real  $x$ .

Let  $g(x) = f^{-1}(x)$ .

Which of the following statements MUST be true for any value of  $x$  where  $f'(x)$  and  $g'(x)$  are both defined?

- A.  $f'(x) = \frac{-1}{g'(x)}$
- B.  $f'(x) = \frac{1}{g'(x)}$
- C.  $f'(x)g'(x) < 0$
- D.  $f'(x)g'(x) > 0$

$f(x)$  and  $g(x)$  are either both monotonically increasing or both monotonically decreasing.

Therefore:

- $f'(x) > 0$  and  $g'(x) > 0$
- $f'(x) < 0$  and  $g'(x) < 0$

Both result in  $f'(x)g'(x) > 0$

- 10 Which statement is always true for real numbers  $a$  and  $b$  where  $0 \leq a < b \leq 1$ ?

- A.  $\cos^{-1} a < \cos^{-1} b$
- B.  $\cos^{-1} a - \sin^{-1} a > \cos^{-1} b - \sin^{-1} b$
- C.  $\cos^2 a < \cos^2 b$
- D.  $\sin a \cos a > \sin b \cos b$

Consider

$$f(x) = \cos^{-1} x - \sin^{-1} x = \cos^{-1} x - \left( \frac{\pi}{2} - \cos^{-1} x \right) = 2 \cos^{-1} x - \frac{\pi}{2}$$

$f(x)$  is a decreasing function for  $0 \leq x \leq 1$  as  $2 \cos^{-1} x$  is a decreasing function.

$$\therefore \cos^{-1} a - \sin^{-1} a > \cos^{-1} b - \sin^{-1} b$$

## Section II

60 marks

Attempt Questions 11-14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

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**Question 11** (15 marks) Use a SEPARATE writing booklet

- (a) The parametric equations of a line are given below.

2

$$x = 2 - 5t$$

$$y = 3t + 1$$

Find the Cartesian equation of this line in the form  $y = mx + c$ .

$$x = 2 - 5t$$

$$t = \frac{2 - x}{5}$$

$$y = 3t + 1$$

$$y = 3\left(\frac{2 - x}{5}\right) + 1$$

$$y = \frac{-3x}{5} + \frac{6}{5} + 1$$

$$y = \frac{-3x}{5} + \frac{11}{5}$$

- (b) Find the inverse function,  $f^{-1}(x)$ , of the function  $f(x) = e^{2x} - 1$ .

2

$$x = e^{2y} - 1$$

$$e^{2y} = x + 1$$

$$2y = \ln(x + 1)$$

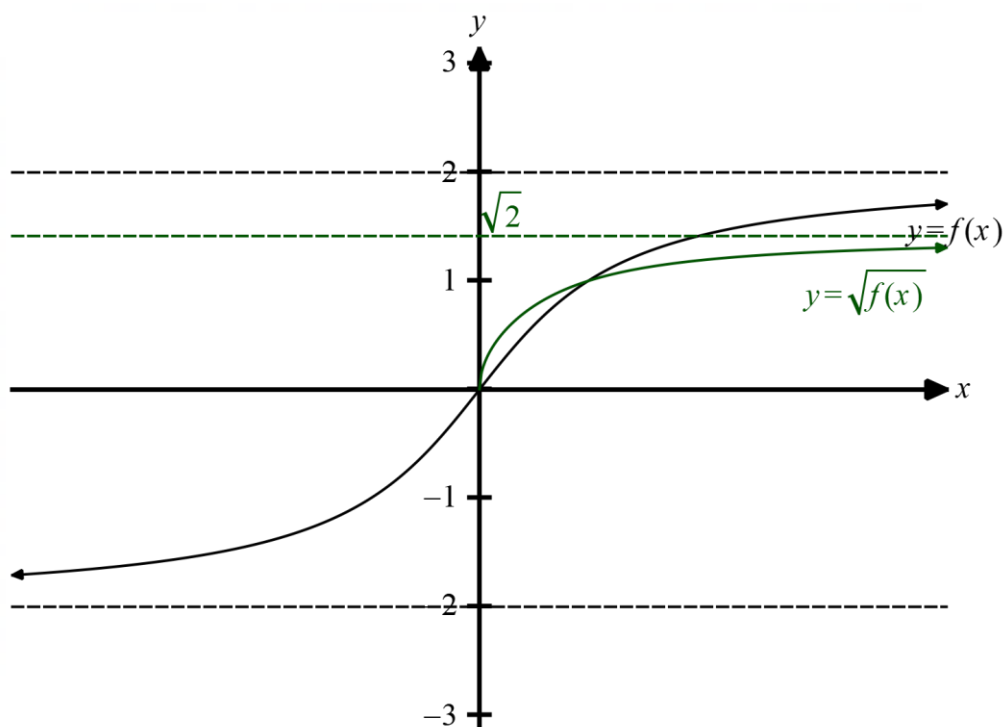
$$y = \frac{1}{2} \ln(x + 1)$$

$$f^{-1}(x) = \frac{1}{2} \ln(x + 1)$$

- (c) The graph of  $y = f(x)$  is given on the separate Response Sheet for Question 11 (c).

2

On the graph provided in the response sheet, sketch the graph of  $y = \sqrt{f(x)}$ , clearly showing all important features.



- (d) Consider the polynomial

3

$$P(x) = x^3 - 2x^2 - 7x - 4.$$

Show that  $x + 1$  is a factor of  $P(x)$  and hence find  $P(x)$  in fully factorised form.

$$P(-1) = (-1)^3 - 2(-1)^2 - 7(-1) - 4 = -1 - 2 + 7 - 4 = 0$$

$$\begin{array}{r} x^2 - 3x - 4 \\ (x+1) \overline{) x^3 - 2x^2 - 7x - 4} \\ \underline{x^3 + x^2} \phantom{- 4} \\ -3x^2 - 7x - 4 \\ \underline{-3x^2 - 3x} \phantom{- 4} \\ -4x - 4 \\ \underline{-4x - 4} \phantom{- 4} \end{array}$$

$$P(x) = (x+1)(x^2 - 3x - 4) = (x+1)(x+1)(x-4) = (x+1)^2(x-4)$$

Question 11 (continued)

(e) Solve  $\frac{2x}{3-x} \leq 4$ .

3

$$\frac{2x}{3-x} \times (3-x)^2 \leq 4 \times (3-x)^2$$

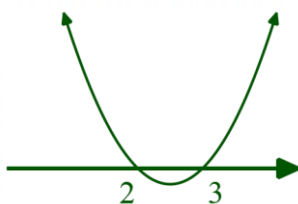
$$2x(3-x) \leq 4(3-x)^2$$

$$4(3-x)^2 - 2x(3-x) \geq 0$$

$$(3-x)[4(3-x) - 2x] \geq 0$$

$$(3-x)(12-6x) \geq 0$$

$$6(x-3)(x-2) \geq 0$$



$$\therefore x \leq 2, x > 3 \quad (\because x \neq 3)$$

(f) Find the coefficient of  $x^{10}$  in the expansion of  $\left(\frac{2}{x} - x^3\right)^6$ .

3

The expansion will contain terms in the form of:

$$\begin{aligned} & \binom{6}{r} \left(\frac{2}{x}\right)^{6-r} (-x^3)^r \\ &= \binom{6}{r} 2^{6-r} x^{r-6} (-1)^r x^{3r} \\ &= \binom{6}{r} 2^{6-r} (-1)^r x^{4r-6} \end{aligned}$$

To obtain  $x^{10}$ :

$$4r - 6 = 10$$

$$r = 4$$

coefficient of  $x^{10}$ :

$$\binom{6}{4} 2^{6-4} (-1)^4 = 60$$

**End of Question 11**

**Question 12** (15 marks) Use a SEPARATE writing booklet

- (a) Use Mathematical induction to prove that  $4n^3 - n$  is divisible by 3 for all integers  $n \geq 1$ . **3**

When  $n = 1$

$$\begin{aligned}4(1)^3 - 1 &= 3 = 3 \times 1 \\ \therefore \text{statement is true for } n = 1\end{aligned}$$

Assume the statement is true for  $n = k$

$$4k^3 - k = 3A \text{ where } A \in \mathbb{Z}$$

When  $n = k + 1$

$$\begin{aligned}&4(k+1)^3 - (k+1) \\&= 4(k^3 + 3k^2 + 3k + 1) - k - 1 \\&= (4k^3 - k) + 3 \times 4k^2 + 3 \times 4k + 4 - 1 \\&= 3A + 3(4k^2 + 4k + 1) \\&= 3(A + 4k^2 + 4k + 1) \\&= 3B \text{ (where } B = A + 4k^2 + 4k + 1 \in \mathbb{Z}, \because A, k \in \mathbb{Z}) \\&\therefore \text{statement is true for } n = k + 1 \text{ if it is true for } n = k.\end{aligned}$$

Hence the statement is true for all integers  $n \geq 1$  by Mathematical Induction.

- (b) (i) Express  $3\sin x - \sqrt{3}\cos x$  in the form  $2\sqrt{3}\sin(x+\alpha)$ , where  $0 < \alpha < 2\pi$ .

2

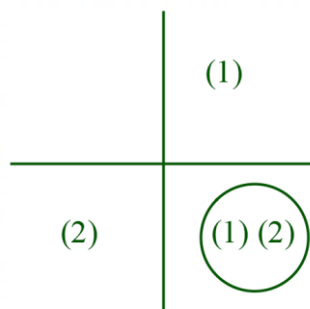
$$3\sin x - \sqrt{3}\cos x = 2\sqrt{3}(\sin x \cos \alpha + \cos x \sin \alpha)$$

$$3\sin x - \sqrt{3}\cos x = (2\sqrt{3}\cos \alpha)\sin x + (2\sqrt{3}\sin \alpha)\cos x$$

By equating coefficients of  $\sin x$  and  $\cos x$ :

$$3 = 2\sqrt{3}\cos \alpha \quad (1) \qquad -\sqrt{3} = 2\sqrt{3}\sin \alpha \quad (2)$$

$$\cos \alpha = \frac{3}{2\sqrt{3}} \quad (1) \qquad \sin \alpha = \frac{-\sqrt{3}}{2\sqrt{3}} \quad (2)$$



$\therefore \alpha$  must be in the 4<sup>th</sup> quadrant

$$\cos \alpha = \frac{3}{2\sqrt{3}}$$

$$\cos \alpha = \frac{\sqrt{3}}{2}$$

$$\alpha = \frac{11\pi}{6}$$

$$3\sin x - \sqrt{3}\cos x = 2\sqrt{3}\sin\left(x + \frac{11\pi}{6}\right)$$

- (ii) Hence, or otherwise, solve  $3\sin x = \sqrt{3}\cos x - \sqrt{3}$ , where  $0 \leq x < 2\pi$ .

2

$$3\sin x = \sqrt{3}\cos x - \sqrt{3}$$

$$3\sin x - \sqrt{3}\cos x = -\sqrt{3}$$

$$2\sqrt{3}\sin\left(x + \frac{11\pi}{6}\right) = -\sqrt{3} \quad (\text{from (i)})$$

$$\sin\left(x + \frac{11\pi}{6}\right) = \frac{-1}{2}$$

$$x + \frac{11\pi}{6} = \frac{11\pi}{6}, \frac{19\pi}{6}$$

$$x = 0, \frac{4\pi}{3}$$

- (c) Evaluate  $\int_0^{\frac{\sqrt{3}}{2}} \frac{8x^2}{\sqrt{1-x^2}} dx$  using the substitution  $x = \sin \theta$ . 4

$$\begin{aligned} x &= \sin \theta \\ \frac{dx}{d\theta} &= \cos \theta && \text{when } x = \frac{\sqrt{3}}{2}, \theta = \frac{\pi}{3} \\ dx &= \cos \theta d\theta && \text{when } x = 0, \theta = 0 \end{aligned}$$

$$\begin{aligned} \int_0^{\frac{\sqrt{3}}{2}} \frac{8x^2}{\sqrt{1-x^2}} dx &= \int_0^{\frac{\pi}{3}} \frac{8 \sin^2 \theta}{\sqrt{1-\sin^2 \theta}} \cdot \cos \theta d\theta = \int_0^{\frac{\pi}{3}} \frac{8 \sin^2 \theta}{\sqrt{\cos^2 \theta}} \cdot \cos \theta d\theta = \int_0^{\frac{\pi}{3}} 8 \sin^2 \theta d\theta \\ \int_0^{\frac{\pi}{3}} 8 \sin^2 \theta d\theta &= 4 \int_0^{\frac{\pi}{3}} 1 - \cos 2\theta d\theta = 4 \left[ \theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{3}} = 4 \left[ \left( \frac{\pi}{3} - \frac{1}{2} \sin \left( 2 \cdot \frac{\pi}{3} \right) \right) - \left( 0 - \frac{1}{2} \sin (2 \cdot 0) \right) \right] \\ &= 4 \left( \frac{\pi}{3} - \frac{\sqrt{3}}{4} \right) \end{aligned}$$

- (d) Charles is a member of an elite racing group of 11 people. A team of 5 people is to be formed from the group. Use a combinatorial argument in this context to prove that 2

$$\binom{11}{5} = \binom{10}{4} + \binom{10}{5}.$$

LHS:

Total number of teams that can be formed by picking 5 people from 11:  $\binom{11}{5}$

RHS:

Case 1: Charles is on the team

Picking the remaining 4 people on the team from the remaining 10 people:  $\binom{10}{4}$

Case 2: Charles is not on the team

Picking the remaining 5 people on the team from the 10 people (excluding Charles):  $\binom{10}{5}$

$$\text{Total: } \binom{10}{4} + \binom{10}{5}$$

(e) By sketching a graph, or otherwise, solve  $(|x|-1)(|x|-2)=6$ .

2

Let  $f(x)=(x-1)(x-2)$

Solutions to  $f(x)=6$ ,

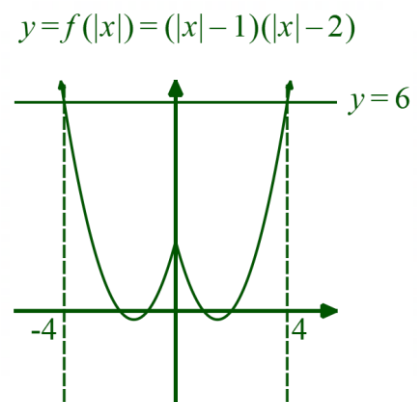
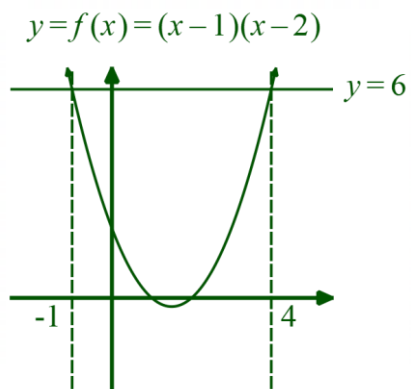
$$(x-1)(x-2)=6$$

$$x^2-3x-4=0$$

$$(x-4)(x+1)=0$$

$$x=-1, 4$$

Consider the graph of  $f(x)=(x-1)(x-2)$



$$\therefore x=-4, 4$$

**End of Question 12**

**Question 13** (15 marks) Use a SEPARATE writing booklet

(a) Let  $f(x) = \tan^{-1} x - \tan^{-1} \left( \frac{x-1}{x+1} \right)$ , for  $x > -1$ .

(i) Show that  $f(x) = k$  where  $k$  is a real number.

2

$$\begin{aligned} f'(x) &= \frac{1}{1+x^2} - \frac{1}{1+\left(\frac{x-1}{x+1}\right)^2} \times \frac{(x+1)(1) - (x-1)(1)}{(x+1)^2} \\ &= \frac{1}{1+x^2} - \frac{2}{(x+1)^2 + (x-1)^2} \\ &= \frac{1}{1+x^2} - \frac{2}{x^2 + 2x + 1 + x^2 - 2x + 1} \\ &= \frac{1}{1+x^2} - \frac{2}{2x^2 + 2} \\ &= \frac{1}{1+x^2} - \frac{1}{1+x^2} \\ &= 0 \end{aligned}$$

$\therefore f'(x) = 0$  then  $f(x)$  must be a constant.  $\therefore f(x) = k$

(ii) Hence, find the value of  $k$ .

1

$$\begin{aligned} f(0) &= \tan^{-1} 0 - \tan^{-1} \left( \frac{0-1}{0+1} \right) = 0 - \tan^{-1}(-1) = 0 - \left( -\frac{\pi}{4} \right) = \frac{\pi}{4} \\ \therefore k &= \frac{\pi}{4} \end{aligned}$$

(b) A particle is moving on the curve with equation

4

$$y = 2 \arcsin 3x, \quad -\frac{1}{3} \leq x \leq \frac{1}{3}.$$

The particle has coordinates  $(x, y)$  at time  $t$ , where distances are measured in metres and time in seconds.

At  $t = T$ , the  $y$  coordinate of the particle is  $\frac{1}{3}\pi$  m and the  $y$  coordinate is increasing at a rate of 2 m/s.

Find the exact rate at which the  $x$  coordinate of the particle is changing at  $t = T$ .

$$\frac{dy}{dx} = 2 \left( \frac{1}{\sqrt{1-(3x)^2}} \right) (3) = \frac{6}{\sqrt{1-9x^2}}$$

when  $y = \frac{1}{3}\pi$

$$\frac{1}{3}\pi = 2 \arcsin 3x$$

$$\arcsin 3x = \frac{\pi}{6}$$

$$3x = \frac{1}{2}$$

$$x = \frac{1}{6}$$

when  $x = \frac{1}{6}$

$$\frac{dy}{dx} = \frac{6}{\sqrt{1-9\left(\frac{1}{6}\right)^2}} = \frac{6}{\sqrt{\frac{3}{4}}} = \frac{12}{\sqrt{3}}$$

$$\frac{dx}{dy} = \frac{\sqrt{3}}{12}$$

At  $t = T$

$$\frac{dx}{dt} = \frac{dx}{dy} \times \frac{dy}{dt} = \frac{\sqrt{3}}{12} \times 2 = \frac{\sqrt{3}}{6} \text{ m/s}$$

- (c) At a particular high school, 17 students, including Xavi and Yuri, are enrolled to study Mathematics Extension 1. There are three Mathematics Extension 1 classes with a maximum class size of 6. Each class is taught by a different teacher.

- (i) Will there always be a class that is full? Justify your answer.

1

$$\frac{\text{number of students}}{\text{number of classes}} = \frac{17}{3} = 5.66... > 5$$

∴ Using the Pigeonhole principle, at least one class has 6 students. There will always be a class that is full.

Or

Distributing 15 students evenly into the 3 classes evenly will give 5 students in each class. The 16<sup>th</sup> must be distributed to a class and make that class full. Therefore, there'll always be a class that is full

- (ii) Students are randomly assigned into classes. Find the number of ways the classes can be assigned to teachers such that Xavi and Yuri are in the same class.

2

The only possible class sizes are 2 classes of 6 students and 1 class of 5 students. Consider two cases:

Case 1: Xavi and Yuri are in a class of 6 students

$${}^{15}C_4 \times {}^{11}C_6 \times {}^5C_5 \times 3! = 3783780$$

Case 2: Xavi and Yuri are in a class of 5 students

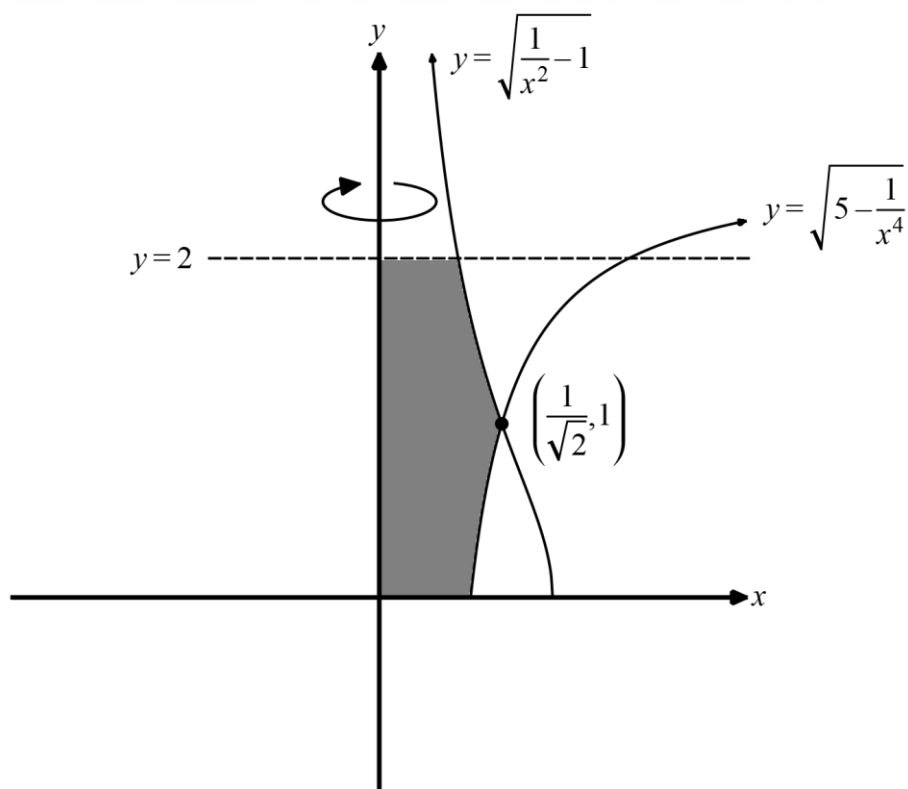
$$\frac{{}^{15}C_3 \times {}^{12}C_6 \times {}^6C_5 \times 3!}{2!} = 1261260$$

Total: 5045040

Question 13 (continued)

- (d) A solid trophy is to be made by rotating the shaded region below around the  $y$ -axis.

5



Show that the volume of the trophy is  $\frac{\pi^2}{4} u^3$ .

$$\begin{aligned} y &= \sqrt{5 - \frac{1}{x^4}} \\ y^2 &= 5 - \frac{1}{x^4} \\ x^4 &= \frac{1}{5 - y^2} \\ x^2 &= \frac{1}{\sqrt{5 - y^2}} \end{aligned}$$

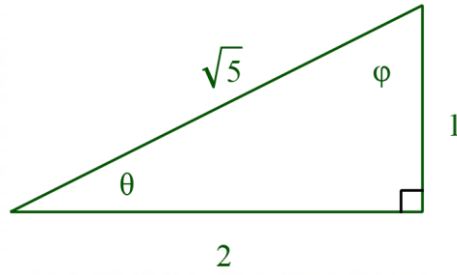
$$\begin{aligned} y &= \sqrt{\frac{1}{x^2} - 1} \\ y^2 &= \frac{1}{x^2} - 1 \\ x^2 &= \frac{1}{1 + y^2} \end{aligned}$$

$$V_1 = \pi \int_0^1 \frac{1}{\sqrt{5-y^2}} dy = \pi \left[ \sin^{-1} \left( \frac{y}{\sqrt{5}} \right) \right]_0^1 = \pi \sin^{-1} \frac{1}{\sqrt{5}}$$

$$V_2 = \pi \int_1^2 \frac{1}{1+y^2} dy = \pi \left[ \tan^{-1} y \right]_1^2 = \pi \left( \tan^{-1} 2 - \tan^{-1} 1 \right) = \pi \left( \tan^{-1} 2 - \frac{\pi}{4} \right)$$

$$V_{total} = \pi \left( \sin^{-1} \frac{1}{\sqrt{5}} + \tan^{-1} 2 - \frac{\pi}{4} \right)$$

Consider the following triangle



$$\theta + \phi = \frac{\pi}{2}$$

$$\sin^{-1} \frac{1}{\sqrt{5}} + \tan^{-1} \frac{2}{1} = \frac{\pi}{2}$$

$$\therefore V_{total} = \pi \left( \sin^{-1} \frac{1}{\sqrt{5}} + \tan^{-1} 2 - \frac{\pi}{4} \right) = \pi \left( \frac{\pi}{2} - \frac{\pi}{4} \right) = \frac{\pi^2}{4} u^3$$

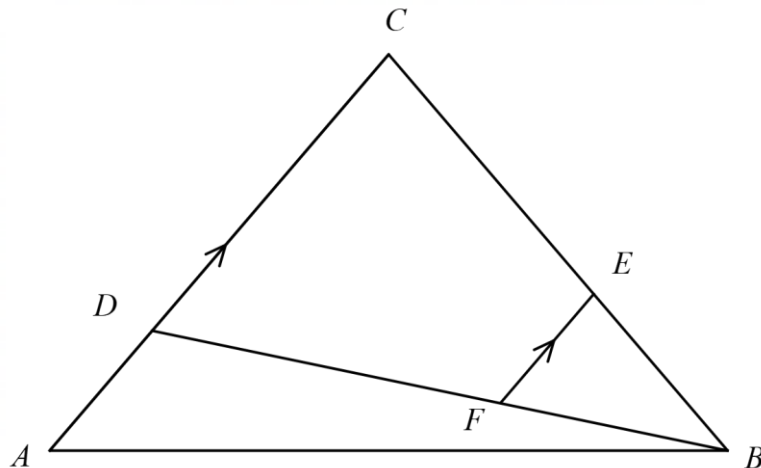
**End of Question 13**

**Question 14** (15 marks) Use a SEPARATE writing booklet

- (a) In triangle  $ABC$ ,  $D$  lies on  $AC$  such that  $\overrightarrow{AD} = \frac{1}{4}\overrightarrow{AC}$ ,  $E$  lies on  $BC$  such that

$$\overrightarrow{BE} = \frac{1}{3}\overrightarrow{BC}.$$

$F$  lies on  $BD$  such that  $FE \parallel AC$ .



- (i) Show that  $\overrightarrow{FE} = \overrightarrow{FB} + \frac{1}{3}\overrightarrow{BC}$ .

1

$$\overrightarrow{FE} = \overrightarrow{FB} + \overrightarrow{BE} = \overrightarrow{FB} + \frac{1}{3}\overrightarrow{BC}$$

- (ii) Let  $\overrightarrow{FE} = k\overrightarrow{DC}$ . Use vector methods to show that  $\overrightarrow{FE} = \overrightarrow{AD}$ .

3

$$k\overrightarrow{DC} = \overrightarrow{FB} + \frac{1}{3}\overrightarrow{BC}$$

$$k(\overrightarrow{DB} + \overrightarrow{BC}) = \overrightarrow{FB} + \frac{1}{3}\overrightarrow{BC}$$

$$k(\lambda\overrightarrow{FB} + \overrightarrow{BC}) = \overrightarrow{FB} + \frac{1}{3}\overrightarrow{BC} \quad (\because \overrightarrow{DB} \parallel \overrightarrow{FB})$$

$$k\lambda\overrightarrow{FB} + k\overrightarrow{BC} = \overrightarrow{FB} + \frac{1}{3}\overrightarrow{BC}$$

$$\therefore k = \frac{1}{3}, \overrightarrow{FE} = \frac{1}{3}\overrightarrow{DC}, \overrightarrow{DC} = 3\overrightarrow{FE}$$

$$\overrightarrow{AD} = \frac{1}{4}(\overrightarrow{AD} + \overrightarrow{DC})$$

$$\frac{3}{4}\overrightarrow{AD} = \frac{1}{4}\overrightarrow{DC}$$

$$3\overrightarrow{AD} = \overrightarrow{DC}$$

$$3\overrightarrow{AD} = 3\overrightarrow{FE}$$

$$\overrightarrow{AD} = \overrightarrow{FE}$$

Question 14 (continued)

- (b) The polynomial  $p(x) = x^3 + bx^2 + cx + d$  has stationary points at  $x = \alpha$  and  $x = \beta$ , where  $\alpha \neq \beta$ .

- (i) Show that  $b^2 > 3c$ .

2

$$p'(x) = 3x^2 + 2bx + c$$

Since there are two stationary points,  $p'(x) = 0$  must have two solutions,  $\therefore \Delta > 0$

$$(2b)^2 - 4(3)(c) > 0$$

$$4b^2 - 12c > 0$$

$$4b^2 > 12c$$

$$b^2 > 3c$$

- (ii) You are given that  $m^3 - n^3 = (m - n)(m^2 + mn + n^2)$ . (Do NOT prove this)

3

Show that  $\frac{p(\alpha) - p(\beta)}{\alpha - \beta} = \frac{2}{9}(3c - b^2)$ .

$$\begin{aligned} p(\alpha) - p(\beta) &= \alpha^3 + b\alpha^2 + c\alpha + d - (\beta^3 + b\beta^2 + c\beta + d) \\ &= \alpha^3 - \beta^3 + b(\alpha^2 - \beta^2) + c(\alpha - \beta) \\ &= (\alpha - \beta)(\alpha^2 + \alpha\beta + \beta^2) + b(\alpha - \beta)(\alpha + \beta) + c(\alpha - \beta) \quad (\text{using the given result}) \\ &= (\alpha - \beta)[(\alpha^2 + \beta^2 + \alpha\beta) + b(\alpha + \beta) + c] \end{aligned}$$

$$\begin{aligned} \frac{p(\alpha) - p(\beta)}{\alpha - \beta} &= (\alpha^2 + \beta^2 + \alpha\beta) + b(\alpha + \beta) + c \\ &= (\alpha + \beta)^2 - \alpha\beta + b(\alpha + \beta) + c \\ &= \left(-\frac{2b}{3}\right)^2 - \left(\frac{c}{3}\right) + b\left(-\frac{2b}{3}\right) + c \\ &= \frac{4b^2}{9} - \frac{2b^2}{3} + \frac{2c}{3} \\ &= \frac{2c}{3} - \frac{2b^2}{9} \\ &= \frac{2}{9}(3c - b^2) \end{aligned}$$

- (iii) Hence, or otherwise, find the relationship between  $\alpha$  and  $\beta$  such that  $p(\alpha) > p(\beta)$ .

1

Method 1: Algebraic using (i) and (ii)

From (i)

$$b^2 > 3c$$

$$3c - b^2 < 0$$

$$\therefore \frac{2}{9}(3c - b^2) < 0$$

RHS of (ii) is negative.

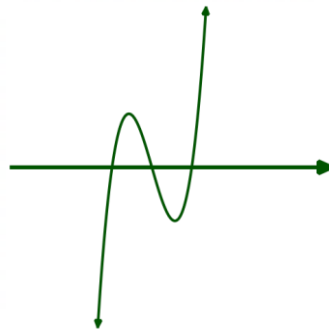
If  $p(\alpha) > p(\beta)$ , then  $p(\alpha) - p(\beta) > 0$ . For LHS of (ii) to also be negative,

$$\alpha - \beta < 0$$

$$\alpha < \beta$$

Method 2: Graphical

$p(x)$  is a monic polynomial of degree 3 with 2 stationary points, its graph must have one maxima and one minima similar to the graph below



If  $p(\alpha) > p(\beta)$ , then  $x = \alpha$  must be the maxima and  $x = \beta$  must be the minima.

$$\therefore \alpha < \beta$$

Question 14 (continued)

- (c) Particle  $A$  is projected from the origin with initial speed  $V \text{ m s}^{-1}$  at an angle  $\theta$  with the horizontal plane. The position vector of particle  $A$ ,  $t$  seconds after it is projected, is given by

$$\vec{r}_A(t) = \begin{pmatrix} Vt \cos \theta \\ Vt \sin \theta - \frac{1}{2}gt^2 \end{pmatrix}. \quad (\text{Do NOT prove this})$$

- (i) Show that the velocity vector of particle  $A$  is given by  $\vec{v}_A(t) = \begin{pmatrix} V \cos \theta \\ V \sin \theta - gt \end{pmatrix}$ . 1

$$\frac{d(\vec{r}_A(t))}{dt} = \frac{d}{dt} \begin{pmatrix} Vt \cos \theta \\ Vt \sin \theta - \frac{1}{2}gt^2 \end{pmatrix}$$

$$\vec{v}_A(t) = \begin{pmatrix} V \cos \theta \\ V \sin \theta - 2\left(\frac{1}{2}gt\right) \end{pmatrix}$$

$$\vec{v}_A(t) = \begin{pmatrix} V \cos \theta \\ V \sin \theta - gt \end{pmatrix}$$

After  $T$  seconds, particle  $B$  is projected from the origin with the same initial speed and angle as particle  $A$ . The position vector of particle  $B$  is given by

$$\vec{r}_B(t) = \begin{pmatrix} V(t-T)\cos\theta \\ V(t-T)\sin\theta - \frac{1}{2}g(t-T)^2 \end{pmatrix}, \quad t \geq T. \quad (\text{Do NOT prove this})$$

- (ii) Show that the particles will never travel in perpendicular directions at any time  $t$  if  $T < \frac{2V\cos\theta}{g}$ .

4

$$\vec{v}_B(t) = \begin{pmatrix} V\cos\theta \\ V\sin\theta - g(t-T) \end{pmatrix}$$

If the particles are travelling at perpendicular directions,  $\vec{v}_A(t) \cdot \vec{v}_B(t) = 0$ .

$$\begin{aligned} (V\cos\theta)(V\cos\theta) + (V\sin\theta - gt)(V\sin\theta - g(t-T)) &= 0 \\ V^2\cos^2\theta + V^2\sin^2\theta - gV\sin\theta(t-T) - gVt\sin\theta + g^2t(t-T) &= 0 \\ V^2 - (gV\sin\theta)t + gVT\sin\theta - gVt\sin\theta + g^2t^2 - (g^2T)t &= 0 \\ g^2t^2 - (2gV\sin\theta + g^2T)t + (V^2 + gVT\sin\theta) &= 0 \end{aligned}$$

The particles will never travel in perpendicular directions if there's no solution to the above quadratic equation in  $t$ .

$$\Delta < 0$$

$$(2gV\sin\theta + g^2T)^2 - 4g^2(V^2 + gVT\sin\theta) < 0$$

$$4g^2V^2\sin^2\theta + 4g^3VT\sin\theta + g^4T^2 - 4g^2V^2 - 4g^3VT\sin\theta < 0$$

$$g^4T^2 < 4g^2V^2 - 4g^2V^2\sin^2\theta$$

$$g^4T^2 < 4g^2V^2(1 - \sin^2\theta)$$

$$T^2 < \frac{4V^2\cos^2\theta}{g^2}$$

$$T < \frac{2V\cos\theta}{g}$$